# Siddharth Naik

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## Summary

Machine learning engineer with 4 years at Virtusa building ML and GenAI systems for enterprise clients: LLM-based issue triage for Google's Gemini API team, a production shipment-delay model on GCP, recommenders and computer-vision prototypes. Mathematical foundation from the Indian Statistical Institute (B.Math, M.Math).

## Technical Skills

**ML & GenAI:** LLMs, RAG, prompt engineering, Gemini API, recommendation systems, classification, regression, clustering, computer vision (SAM, BEiT)

**Frameworks:** PyTorch, TensorFlow, Hugging Face, scikit-learn, XGBoost, LangChain, FAISS, Pandas, Polars, FastAPI, Streamlit

**Cloud & MLOps:** GCP (Vertex AI, Kubeflow Pipelines, BigQuery, Cloud Run, Document AI), Azure Databricks, Docker, MLflow

**Languages & Tools:** Python, SQL, Cypher (Neo4j), JavaScript, Bash, Git

## Experience

**Virtusa, Hyderabad (Remote)**

Senior Software Engineer, Data & Analytics — Apr 2024 – Present

Software Engineer, Data & Analytics — Jun 2022 – Mar 2024

- **LLM Issue Triage Automation** — *Google: Gemini API, AI Studio, Antigravity (vendor team)*
  - Built Gemini-based tooling that routes incoming developer issues, generates channel metadata, drafts replies, and writes reproduction steps and root-cause notes for escalations; cut batch triage time by 30% (measured with vs. without the tools).
  - Led a vendor team that grew from 3 to 6 engineers; triaged 4,000+ issues across the public forum and Google's internal tracker, and reproduced and escalated 200+ unique bugs to Google engineering. Now extending the automation to Google's internal AI developer tools.
- **Shipment Transit-Time Prediction** — *US home-furnishings retailer*
  - Led a 4-person team that took an XGBoost transit- and loading-time model from POC to production on 2 years of shipment data, cutting error from ∼6 days (client's existing model) to a 24-hour RMSE with 10% MAPE.
  - Built the GCP pipeline end to end on Kubeflow (SQL ingestion, training, evaluation, weekly batch inference in Docker, RMSE and MAPE monitoring) and a Streamlit dashboard on Cloud Run that operations used to flag chronically late vendors.
- **Market Basket Analysis** — *US home-furnishings retailer*
  - Built a monthly Kubeflow pipeline mining association rules (mlxtend H-Mine) across 80,000 prioritised transaction filters over 2 years of sales, writing insights to BigQuery; moving processing from Pandas to Polars cut multi-day runs to hours.
  - Shipped a Streamlit dashboard and a serverless service that computes and caches insights for new filters on demand, used to guide product placement and recommendations.
- **Computer Vision & Document AI Prototypes** — *Global records-management company*
  - Built a drone-footage damage-detection POC: SAM segmentation with rule-based filtering to isolate boxes and a fine-tuned BEiT classifier (0.90 recall), 30× faster than manual inspection; a FastAPI service turned detections into reboxing requests in the client's SKU system. The POC informed a fiscal-year-end infrastructure decision.
  - Built a schema-driven document-extraction POC with Google Document AI and OCR, demonstrating data entry about 10× faster than manual keying.
- **Recommendation & Sales-Propensity Models** — *Global medical-equipment manufacturer*
  - Built a hybrid recommender on Azure Databricks: implicit ALS on SME-weighted purchase ratings, adjusted with customer and product clusters to extend recommendations to new products (MRR 0.20).
  - Built a Random Forest model scoring the probability that a sales pitch converts (∼80% accuracy), used by the sales team to prioritise accounts.
- **Innovation & Hiring** — *Internal*
  - Top 3 of 300+ entries, Virtusa Icon 2024: built a Gemini 1.5 physician assistant that turns consultation transcripts into prescriptions, schedules and ICD/CPT-coded notes (FAISS retrieval), estimated to save up to 10 minutes per visit.
  - Conducted 50+ technical interviews for MLOps and AI roles; built rapid prototypes (RAG insurance bot, graph RAG on Neo4j, cattle-weight vision model) and technical decks for RFPs.

## Education

**Indian Statistical Institute, Kolkata**

Master of Mathematics (M.Math), 74% — Dec 2020 – Jun 2022

**Indian Statistical Institute, Bangalore**

Bachelor of Mathematics (B.Math Hons.), 87% — Jul 2017 – Jul 2020

**WorldQuant University (online, part-time)**

MSc in Financial Engineering — Oct 2024 – Present

## Awards

**Young Innovator Award:** for designing and demonstrating GenAI applications — 2024